

Lesson 12: Introduction to Hypothesis Testing

BSTA 551: Statistical Inference

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Review: Where We Left Off

Lessons 9-11: Bootstrap Methods and Confidence Intervals

- Bootstrap principle: resample from sample to estimate sampling distribution
- Bootstrap percentile and t intervals
- Relationship between confidence intervals and plausible parameter values

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Today's Goals:

1. Understand the hypothesis testing framework
2. Define null and alternative hypotheses
3. Learn about test statistics and rejection regions
4. Understand Type I and Type II errors
5. Apply these concepts to medical research examples

From Estimation to Testing

Confidence Intervals

- “What is a plausible range for the parameter?”
- Example: “95% CI for mean reduction in blood pressure is (8, 16) mmHg”
- Provides a range of values

Hypothesis Testing

- “Is there evidence that the parameter differs from a specific value?”
- Example: “Does this drug reduce blood pressure at all?”
- Provides a yes/no decision

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Key Insight (Devore 9.1)

Hypothesis testing is about deciding which of two contradictory claims about a parameter is correct based on sample data.

The Hypothesis Testing Framework

What is a Statistical Hypothesis?

! Definition (Devore 9.1)

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Examples:

- The claim $\mu = 120$ mmHg, where μ is the true average systolic blood pressure
- The statement $p < 0.10$, where p is the proportion of patients who experience adverse events
- The assertion $\mu_{\text{treatment}} - \mu_{\text{control}} > 0$

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! Definitions (Devore 9.1)

Null Hypothesis (H_0): The claim that is initially assumed to be true (the “prior belief” or “status quo” claim).

Alternative Hypothesis (H_a or H_A): The assertion that is contradictory to H_0 — what we might conclude if the data provides strong evidence against H_0 .

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Alternative Hypothesis (H_a or H_A): The assertion that is contradictory to H_0 — what we might conclude if the data provides strong evidence against H_0 .

The null hypothesis will be rejected in favor of the alternative **only if** sample evidence strongly suggests that H_0 is false.

The Courtroom Analogy

Criminal Trial (US)

- **Initial assumption:** Defendant is innocent
- **Burden of proof:** On prosecution
- **Standard:** “Beyond reasonable doubt”
- **Decision:** Guilty or Not Guilty

Note: “Not Guilty” $\neq$ “Innocent”

Hypothesis Testing

- **Initial assumption:** H_0 is true
- **Burden of proof:** On data to contradict H_0
- **Standard:** Significance level α
- **Decision:** Reject H_0 or Fail to Reject H_0

Note: “Fail to Reject” $\neq$ “ H_0 is True”

Visualizing the Analogy

► Code

The Hypothesis Testing Analogy

	Initial Belief	Evidence	Decision 1	Decision 2
Statistics	H_0 assumed true	Sample data collected	Reject H_0 (favor H_a)	Fail to reject H_0 (H_0 plausible)
Courtroom	Innocent until proven guilty	Prosecution presents evidence	Guilty (reject innocence)	Not Guilty (fail to reject)

Two Possible Conclusions

Important Distinction

A hypothesis test can only lead to one of two conclusions:

1. **Reject H_0** — Evidence strongly suggests H_0 is false
2. **Fail to reject H_0** — Evidence does not strongly contradict H_0

We **never** say “Accept H_0 ” because failing to reject is not the same as proving H_0 is true!

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Analogy: A “Not Guilty” verdict doesn’t prove innocence — it means there wasn’t enough evidence to convict.

Formulating Hypotheses

Example 1: New Antihypertensive Drug

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$$H_0 : \mu = 10 \quad (\text{no improvement over standard})$$

$$H_a : \mu > 10 \quad (\text{new drug is better})$$

Example 1: New Antihypertensive Drug

A pharmaceutical company has developed a new drug to lower blood pressure. The current standard treatment has a mean reduction of 10 mmHg. Let μ denote the true mean BP reduction with the new drug.

Question: Is the new drug better than the standard?

Hypotheses:

$$H_0 : \mu = 10 \quad (\text{no improvement over standard})$$

$$H_a : \mu > 10 \quad (\text{new drug is better})$$

The company will only switch to the new drug if there is **convincing evidence** of improvement.

Example 2: Is Normal Body Temperature Really 98.6°F?

The widely accepted “normal” human body temperature of 98.6°F (37°C) dates back to a 19th century German study. Recent research suggests this value may be inaccurate.

Let μ denote the true mean body temperature of healthy adults.

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$$H_a : \mu \neq 98.6 \quad (\text{traditional value is wrong — either direction})$$

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Hypotheses:

$$H_0 : \mu = 98.6 \quad (\text{traditional value is correct})$$

$$H_a : \mu \neq 98.6 \quad (\text{traditional value is wrong — either direction})$$

This is a **two-sided** test because departures in either direction would be scientifically important.

Example 3: Adverse Event Rate

An investigational drug is considered safe if the rate of serious adverse events is at most 5%. In a clinical trial, researchers want to determine if the drug exceeds this threshold.

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Hypotheses:

$$H_0 : p = 0.05 \quad (\text{drug meets safety threshold})$$

$$H_a : p > 0.05 \quad (\text{drug exceeds threshold — unsafe})$$

This is a **one-sided** (upper-tailed) test because only a rate higher than 5% would be concerning.

General Structure of Hypotheses

The null hypothesis is typically stated as an **equality**:

$$H_0 : \theta = \theta_0$$

where θ_0 is called the **null value** of the parameter.

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The alternative hypothesis takes one of three forms:

Alternative	Test Type	Example Context
$H_a : \theta > \theta_0$	Upper-tailed	New treatment better than standard
$H_a : \theta < \theta_0$	Lower-tailed	Reduced error rate
$H_a : \theta \neq \theta_0$	Two-tailed	Testing for any difference

Choosing the Alternative: Research vs. Null

Practical Guidance (Devore 9.1)

- H_a is often called the **research hypothesis** — what the investigator wants to demonstrate
- H_0 represents the **status quo** or “no effect/no difference”
- The burden of proof is on showing H_a is true

This is **conservative**: we only claim an effect when the data provides strong evidence.

Choosing the Alternative: Research vs. Null

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Medical Research Context:

- New treatment must **prove** it's better than standard
- We protect against false claims of efficacy
- FDA approval requires strong evidence that benefits outweigh risks

Test Procedures

Components of a Test Procedure

! Definition (Devore 9.1)

A **test procedure** is specified by:

1. A **test statistic**: a function of the sample data used to make the decision
2. A **rejection region** (critical region): the set of test statistic values for which H_0 will be rejected

The null hypothesis is rejected if and only if the observed test statistic value falls in the rejection region.

Example: Testing Vaccine Efficacy

A new vaccine is tested. Let p be the true proportion of vaccinated individuals who develop the disease. We test $H_0 : p = 0.20$ vs $H_a : p < 0.20$.

In a trial of $n = 200$ vaccinated individuals:

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In a trial of $n = 200$ vaccinated individuals:

Test Statistic: X = number who develop disease

Rejection Region: Reject H_0 if $X \leq 30$

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```
1 # Under H0: X ~ Binomial(200, 0.20)
2 # Expected value under H0
3 n <- 200
4 p0 <- 0.20
5 expected_X <- n * p0
6 cat("Expected X under H0:", expected_X, "\n")
```

Expected X under H0: 40

```
1 cat("Rejection region: X ≤ 30 (substantially less than 40)")
```

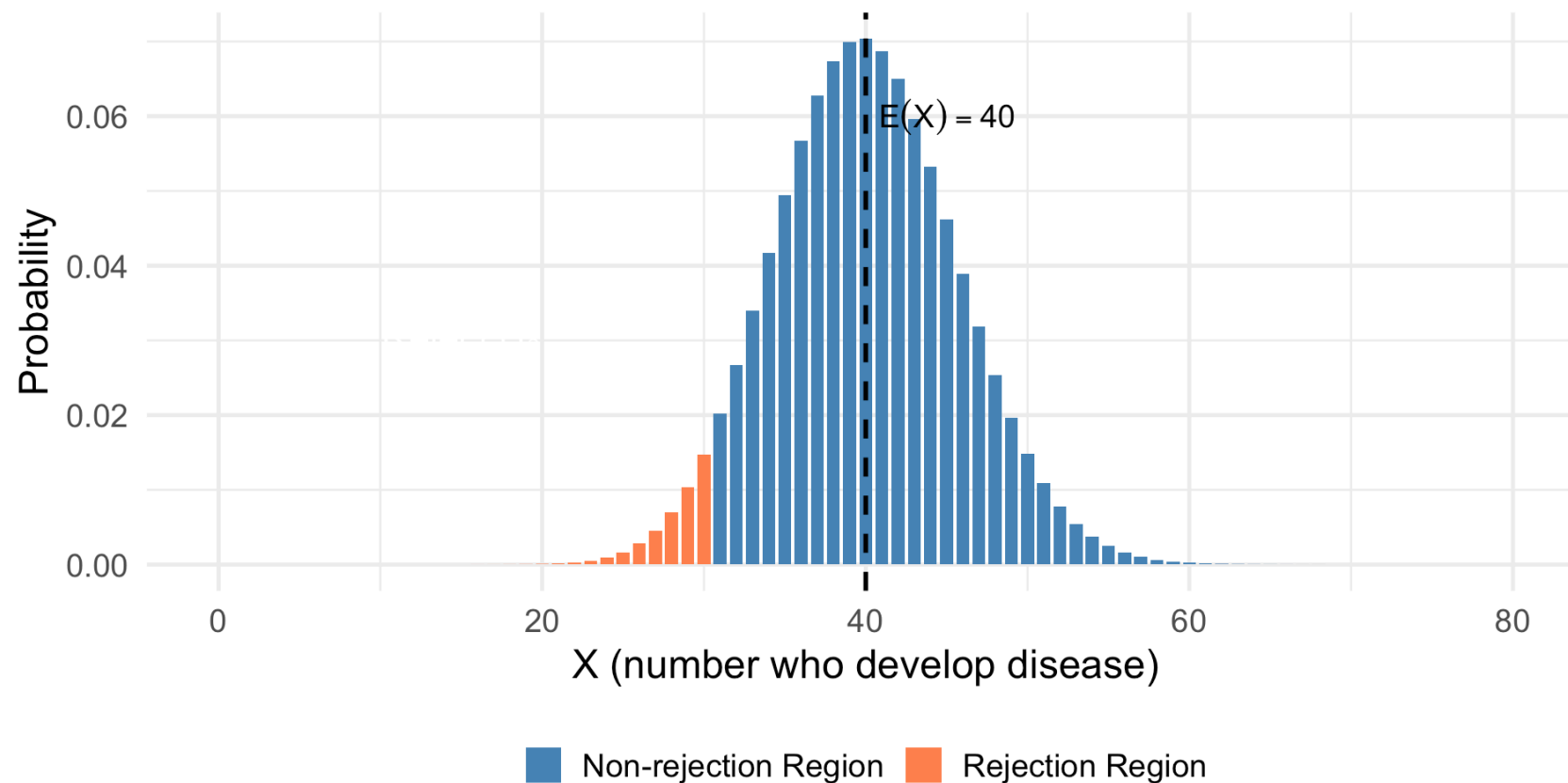
Rejection region: $X \leq 30$ (substantially less than 40)

Visualizing the Rejection Region

► Code

Distribution of X under $H_0: p = 0.20$

Rejection region: $X \leq 30$



Example: Testing Engine Fuel Efficiency (Devore Example 9.3)

A manufacturer considers modifying truck engines to improve fuel efficiency. Current mean: 25 mpg, SD: 3 mpg (known). Let μ = true mean mpg for modified engines.

$$H_0 : \mu = 25 \quad \text{vs} \quad H_a : \mu > 25$$

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Test with $n = 10$ trucks:

Test Statistic: $\bar{X}$ = sample mean fuel efficiency

Possible Rejection Region: Reject H_0 if $\bar{X} \geq 27.5$

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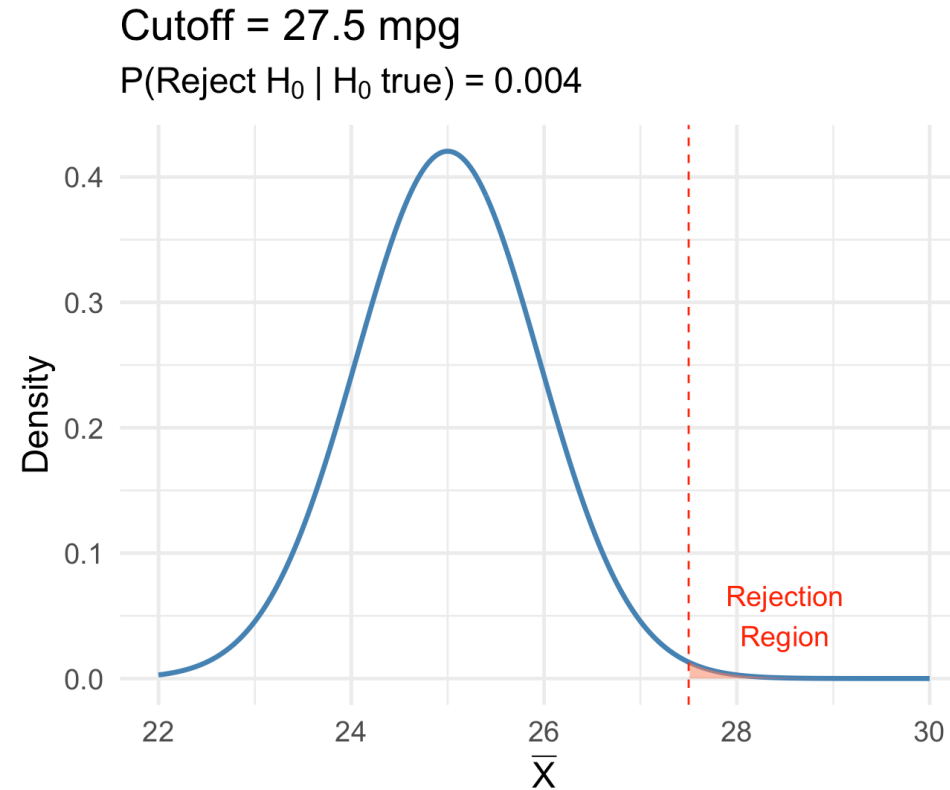
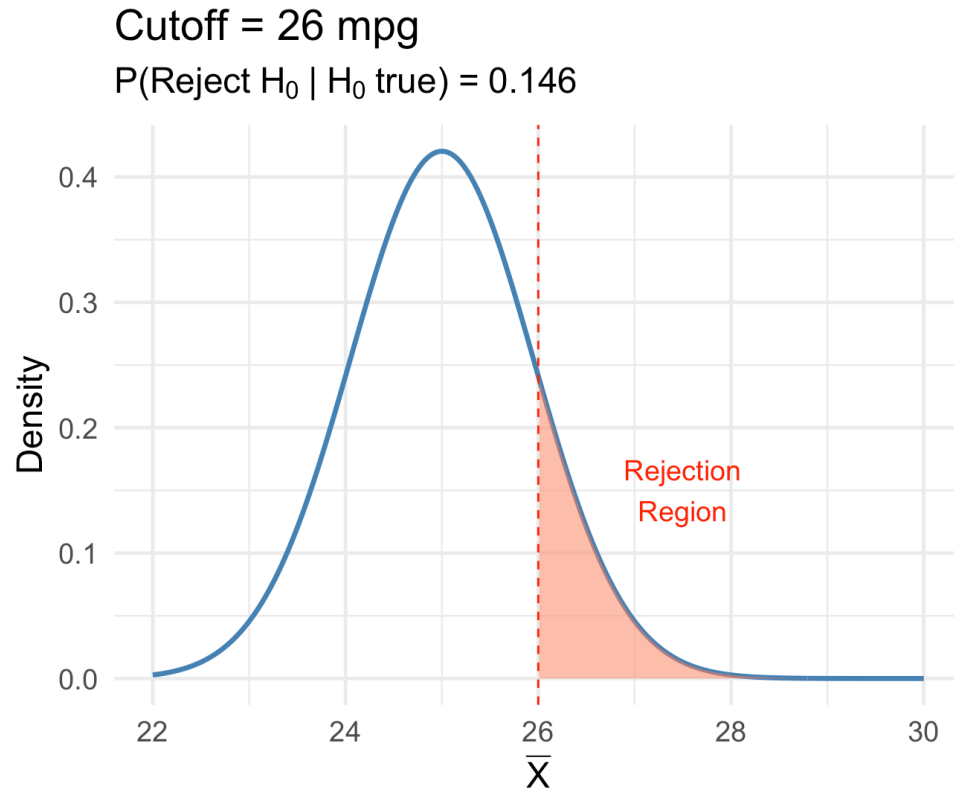
Test Statistic: $\bar{X}$ = sample mean fuel efficiency

Possible Rejection Region: Reject H_0 if $\bar{X} \geq 27.5$

Key Question: How do we choose the cutoff value (27.5 vs 26 vs 28)?

The Tradeoff in Choosing the Cutoff

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Lower cutoff: More likely to reject H_0 (even when true!)

Higher cutoff: Less likely to reject H_0 (even when false!)

Type I and Type II Errors

Two Types of Errors

When making a decision based on sample data, two types of errors are possible:

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	H_0 True	H_0 False
Reject H_0	Type I Error	Correct Decision
Fail to Reject H_0	Correct Decision	Type II Error

Error Probabilities

! Notation

$$\alpha = P(\text{Type I Error}) = P(\text{Reject } H_0 \mid H_0 \text{ is true})$$

$$\beta = P(\text{Type II Error}) = P(\text{Fail to reject } H_0 \mid H_0 \text{ is false})$$

$$\text{Power} = 1 - \beta = P(\text{Reject } H_0 \mid H_0 \text{ is false})$$

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- α is called the **significance level** of the test
- Common choices: $\alpha = 0.05$, $\alpha = 0.01$, $\alpha = 0.10$

Medical Context: Errors Have Consequences

Example: Testing whether a new cancer drug is effective

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Type I Error (False Positive):

- Conclude drug works when it doesn't
- Patients receive ineffective treatment
- Resources wasted on worthless drug
- *Consequence:* Harm to patients, wasted resources

Medical Context: Errors Have Consequences

Example: Testing whether a new cancer drug is effective

Type I Error (False Positive):

- Conclude drug works when it doesn't
- Patients receive ineffective treatment
- Resources wasted on worthless drug
- *Consequence:* Harm to patients, wasted resources

Type II Error (False Negative):

- Conclude drug doesn't work when it does
- Effective treatment never reaches patients
- *Consequence:* Missed opportunity to save lives

Example: Drug Approval Errors (Devore Example 9.4)

Scenario: Testing a new antihypertensive drug against standard care.

$H_0 : \mu_{\text{new}} = \mu_{\text{standard}}$ (no improvement)

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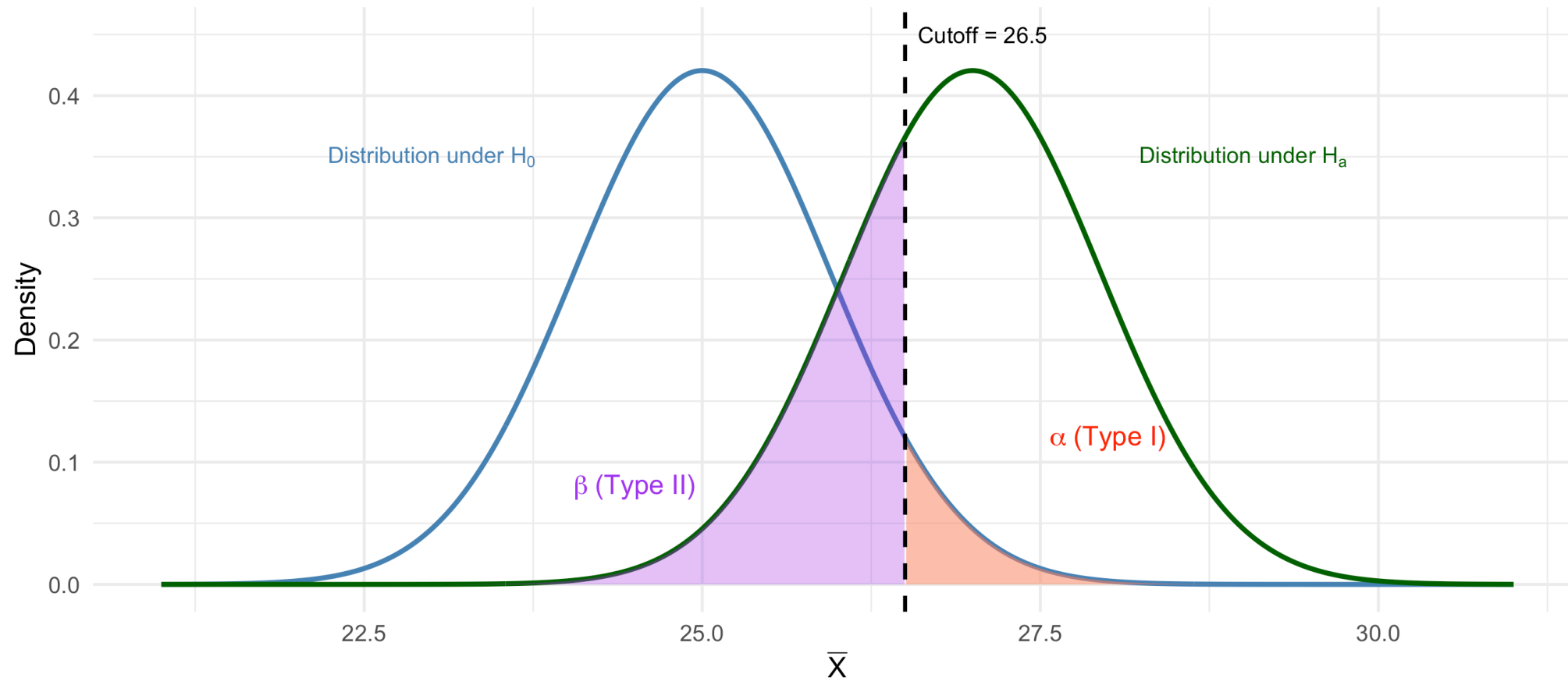
FDA Approach: Keep α small (typically 0.05) — protect against approving ineffective drugs.

Visualizing Type I and Type II Errors

► Code

Type I and Type II Errors Visualized

Fuel efficiency example: $H_0: \mu = 25$ vs $H_a: \mu > 25$



Computing Error Probabilities: Example

For the fuel efficiency test with cutoff $\bar{x} \geq 27.5$:

```
1 mu0 <- 25
2 sigma <- 3
3 n <- 10
4 se <- sigma / sqrt(n)
5 cutoff <- 27.5
6
7 # Type I error: P(reject H0 | H0 true)
8 alpha <- 1 - pnorm(cutoff, mean = mu0, sd = se)
9 cat("α = P( $\bar{X} \geq", cutoff, " | \mu =", mu0, ") =", round(alpha, 4), "\n")$ 
```

$\alpha = P(\bar{X} \geq 27.5 \mid \mu = 25) = 0.0042$

```
1 # Type II error: P(fail to reject | H0 false)
2 # Suppose true mean is  $\mu = 26.5$ 
3 mu_true <- 26.5
4 beta <- pnorm(cutoff, mean = mu_true, sd = se)
5 cat("β = P( $\bar{X} <", cutoff, " | \mu =", mu_true, ") =", round(beta, 4), "\n")$ 
```

$\beta = P(\bar{X} < 27.5 \mid \mu = 26.5) = 0.8541$

```
1 cat("Power = 1 - β =", round(1 - beta, 4))
```

Power = 1 - β = 0.1459

The Fundamental Tradeoff

The Tradeoff (Devore 9.1)

For a **fixed sample size**, decreasing α (Type I error) necessarily **increases** β (Type II error), and vice versa.

The only way to decrease **both** error probabilities simultaneously is to **increase the sample size**.

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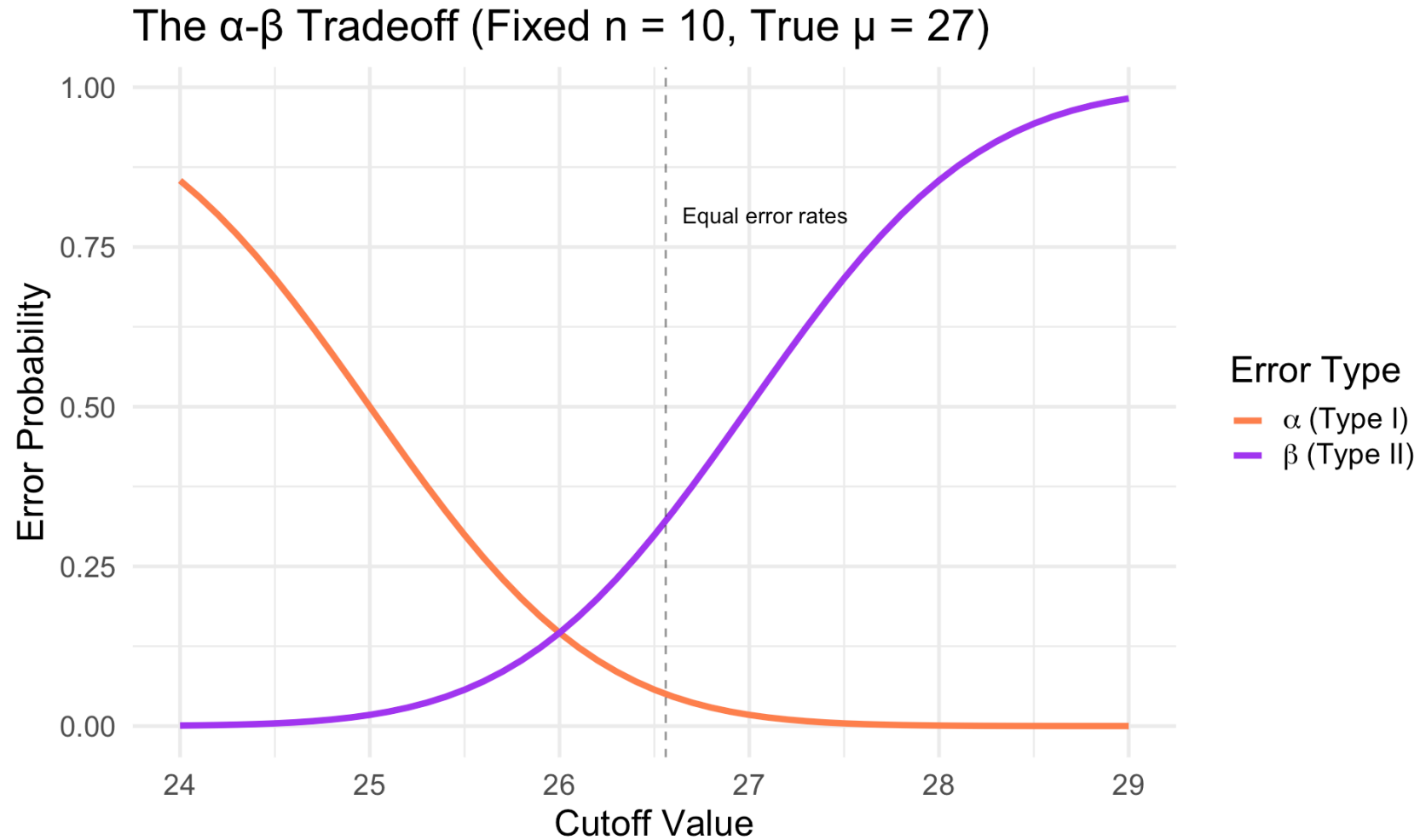
```
1 # Demonstrate tradeoff with different cutoffs
2 cutoffs <- c(26, 26.5, 27, 27.5, 28)
3 alpha_vals <- 1 - pnorm(cutoffs, mu0, se)
4 beta_vals <- pnorm(cutoffs, mean = 27, sd = se) # assuming true mu = 27
5
6 tibble(Cutoff = cutoffs, Alpha = round(alpha_vals, 4), Beta = round(beta_vals, 4)) |>
7   mutate(Power = 1 - Beta)
```

A tibble: 5 × 4

	Cutoff	Alpha	Beta	Power
	<dbl>	<dbl>	<dbl>	<dbl>
1	26	0.146	0.146	0.854
2	26.5	0.0569	0.299	0.701
3	27	0.0175	0.5	0.5
4	27.5	0.0042	0.701	0.299
5	28	0.0008	0.854	0.146

Visualizing the Tradeoff

► Code



Simulation: Demonstrating Error Types

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Simulated Type I Error Rate: 0.061

► Code

Theoretical α : 0.0569

► Code

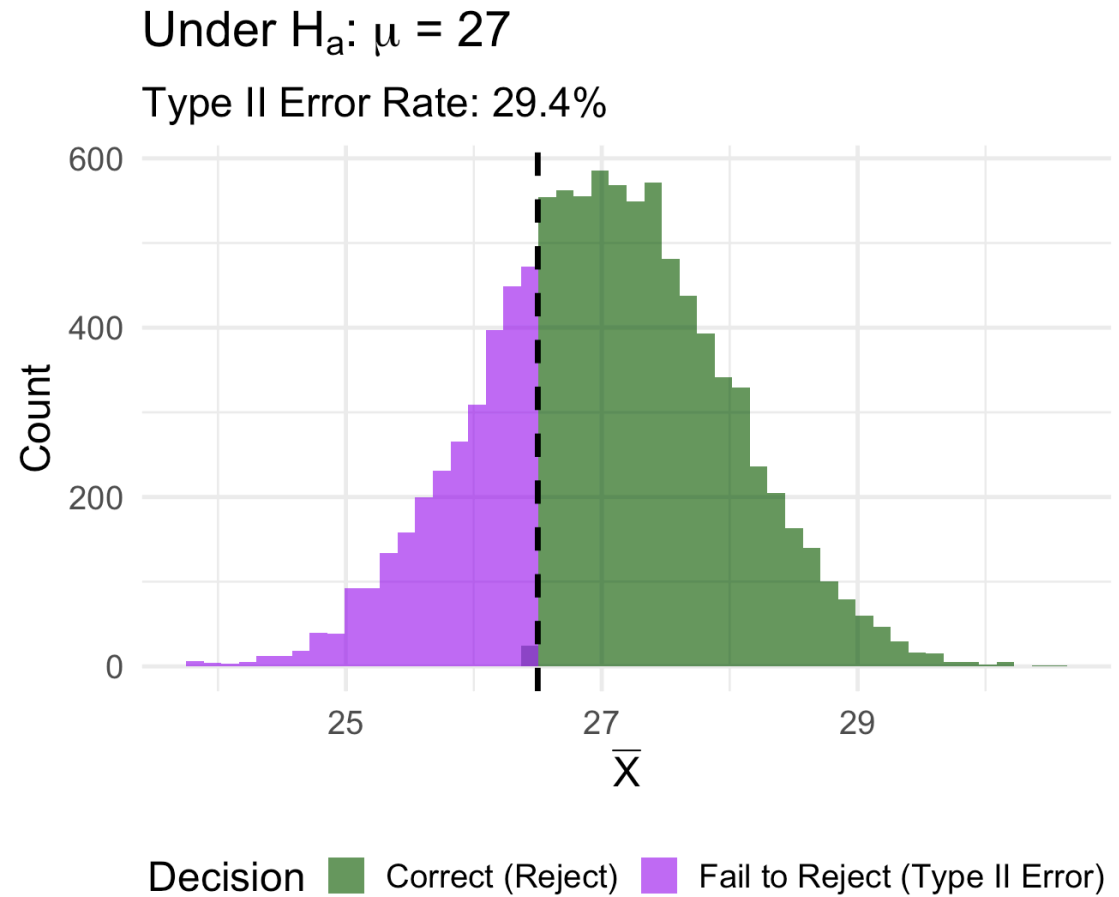
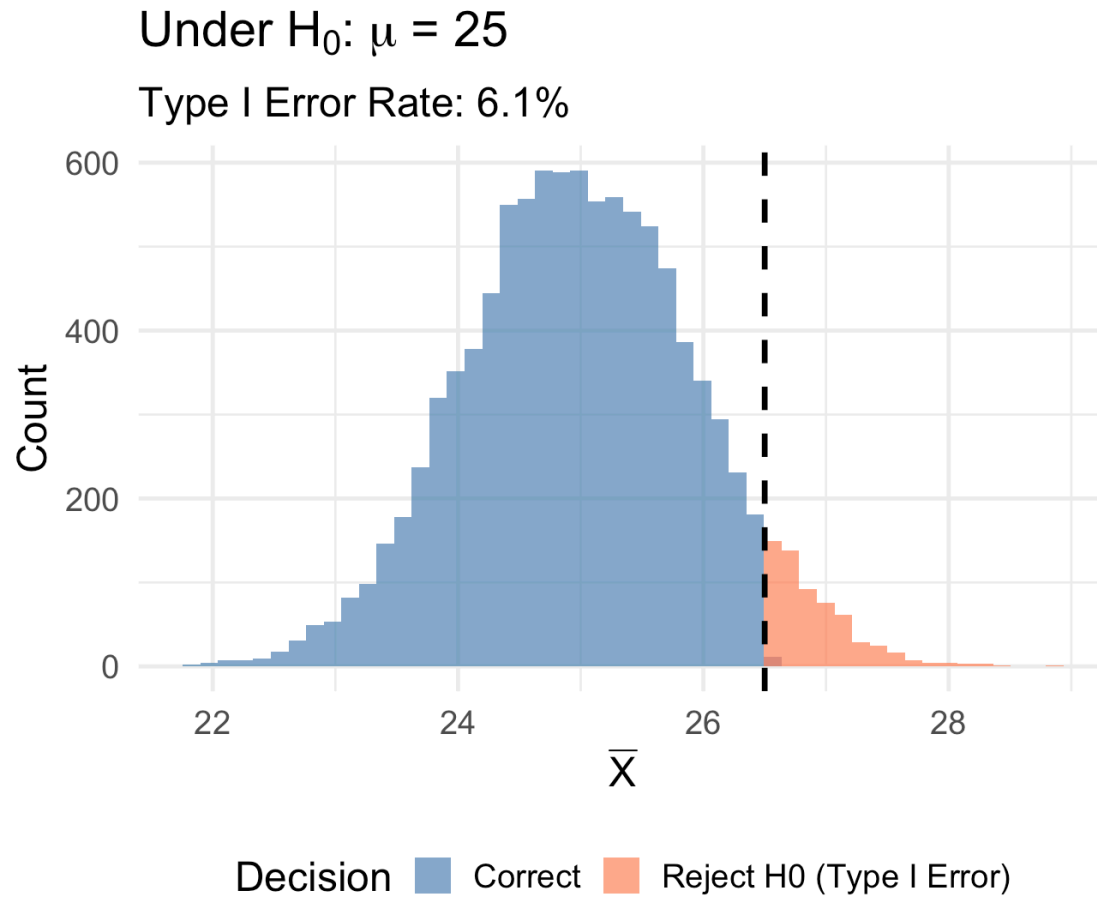
Simulated Type II Error Rate: 0.2939

► Code

Theoretical β : 0.2991

Visualizing Simulated Decisions

► Code



Significance Level and Rejection Regions

The Significance Level

! Definition

The **significance level** α of a test is the probability of making a Type I error — rejecting H_0 when it is actually true.

$$\alpha = P(\text{Reject } H_0 \mid H_0 \text{ is true})$$

Common choices: $\alpha = 0.05$, $\alpha = 0.01$, $\alpha = 0.10$

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Common choices: $\alpha = 0.05$, $\alpha = 0.01$, $\alpha = 0.10$

Why prioritize controlling Type I error?

- Type I errors can lead to **false claims** (e.g., ineffective drugs approved)
- Scientific community uses $\alpha = 0.05$ as a standard threshold
- α is chosen **before** collecting data

Constructing Rejection Regions

Once α is chosen, we determine the rejection region:

Upper-tailed test ($H_a : \mu > \mu_0$):

$$\text{Reject } H_0 \text{ if } \bar{X} \geq \mu_0 + z_\alpha \cdot \frac{\sigma}{\sqrt{n}}$$

Lower-tailed test ($H_a : \mu < \mu_0$):

$$\text{Reject } H_0 \text{ if } \bar{X} \leq \mu_0 - z_\alpha \cdot \frac{\sigma}{\sqrt{n}}$$

Two-tailed test ($H_a : \mu \neq \mu_0$):

$$\text{Reject } H_0 \text{ if } |\bar{X} - \mu_0| \geq z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

Example: Calculating Critical Values

For the fuel efficiency test at $\alpha = 0.05$:

```
1 alpha <- 0.05
2 mu0 <- 25
3 sigma <- 3
4 n <- 10
5 se <- sigma / sqrt(n)
6
7 # Critical value for upper-tailed test
8 z_alpha <- qnorm(1 - alpha)
9 critical_value <- mu0 + z_alpha * se
10
11 cat("z_α =", round(z_alpha, 3), "\n")
```

$z_\alpha = 1.645$

```
1 cat("Critical value:  $\bar{X} \geq$ ", round(critical_value, 3), "\n")
```

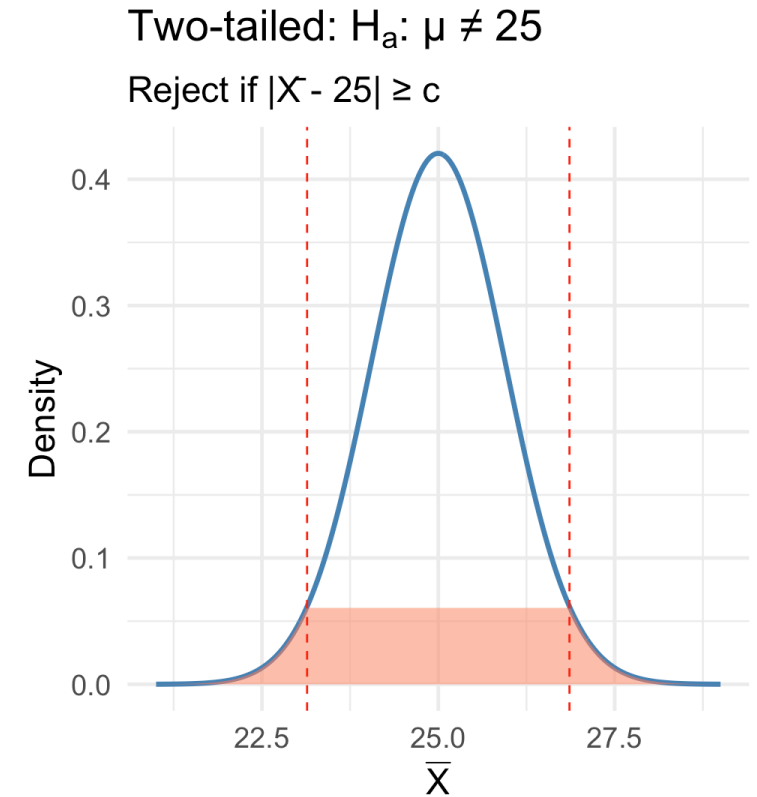
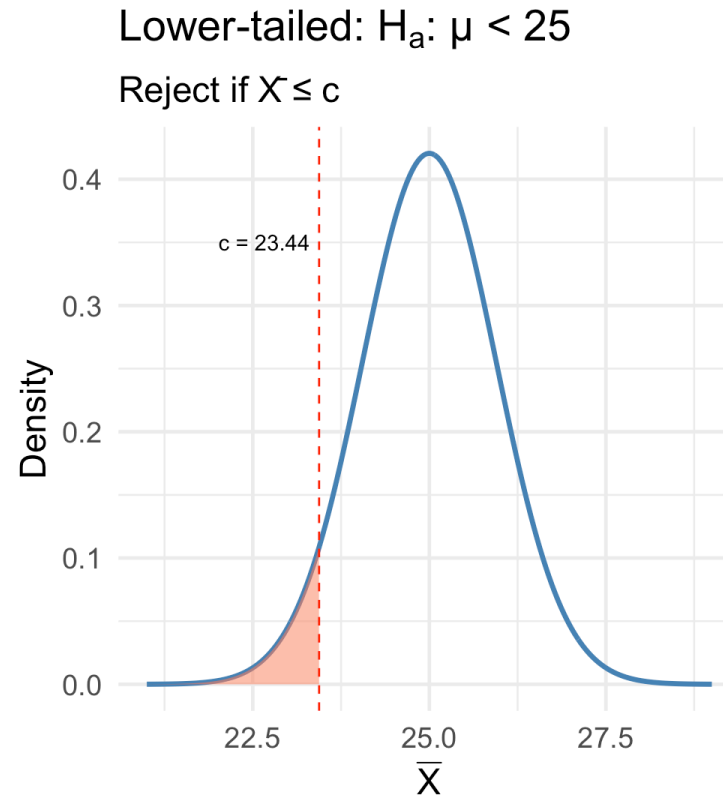
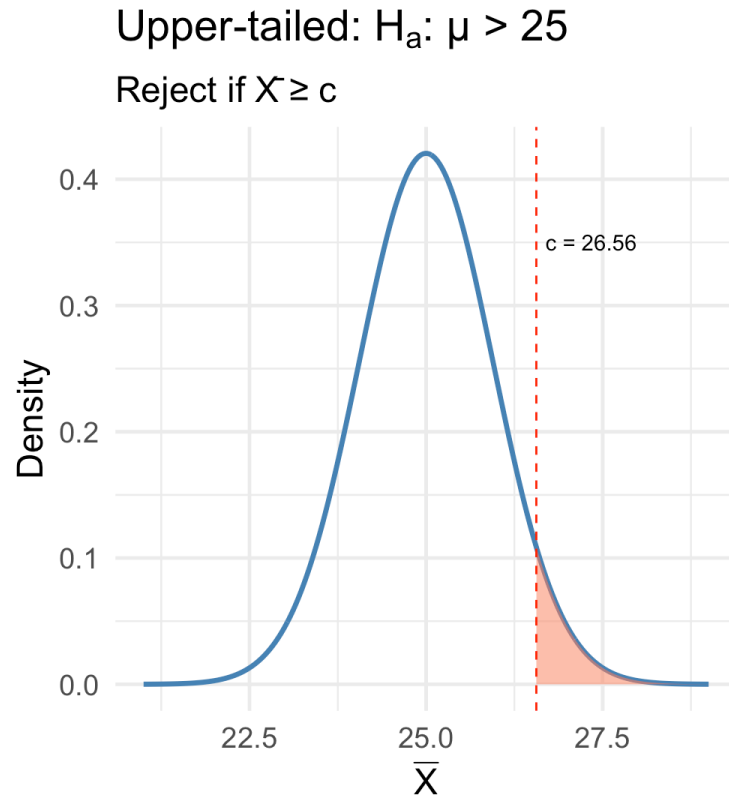
Critical value: $\bar{X} \geq 26.56$

```
1 cat("Rejection region:  $\bar{X} \geq$ ", round(critical_value, 3))
```

Rejection region: $\bar{X} \geq 26.56$

Visualizing Critical Regions by Tail Type

► Code



Medical Research Example: Clinical Trial

Complete Example: Blood Pressure Drug Trial

Setting: Testing a new antihypertensive drug.

Data: $n = 25$ patients, mean BP reduction $\bar{x} = 12.5$ mmHg, known population SD $\sigma = 8$ mmHg.

Question: Is there evidence that the true mean reduction exceeds 10 mmHg?

Complete Example: Blood Pressure Drug Trial

Setting: Testing a new antihypertensive drug.

Data: $n = 25$ patients, mean BP reduction $\bar{x} = 12.5$ mmHg, known population SD $\sigma = 8$ mmHg.

Question: Is there evidence that the true mean reduction exceeds 10 mmHg?

Step 1: State Hypotheses

$$H_0 : \mu = 10 \quad \text{vs} \quad H_a : \mu > 10$$

Step 2: Choose significance level

$$\alpha = 0.05$$

Step 3: Identify Test Statistic and Rejection Region

```
1 # Given data
2 xbar <- 12.5
3 sigma <- 8
4 n <- 25
5 mu0 <- 10
6 alpha <- 0.05
7
8 # Standard error
9 se <- sigma / sqrt(n)
10
11 # Test statistic:  $Z = (\bar{X} - \mu_0) / (\sigma/\sqrt{n})$ 
12 z_stat <- (xbar - mu0) / se
13 cat("Test statistic: z =", round(z_stat, 3), "\n")
```

Test statistic: z = 1.562

```
1 # Critical value for  $\alpha = 0.05$  (upper-tailed)
2 z_crit <- qnorm(1 - alpha)
3 cat("Critical value: z_α =", round(z_crit, 3), "\n")
```

Critical value: z_α = 1.645

```
1 # Rejection region
2 cat("Rejection region: z ≥", round(z_crit, 3))
```

Rejection region: z ≥ 1.645



Step 4: Make Decision

```
1 # Decision
2 if (z_stat >= z_crit) {
3   cat("Decision: Reject H0\n\n")
4   cat("Interpretation: There is statistically significant evidence at  $\alpha = 0.05$ \n")
5   cat("that the true mean BP reduction exceeds 10 mmHg.")
6 } else {
7   cat("Decision: Fail to reject H0\n\n")
8   cat("Interpretation: There is insufficient evidence at  $\alpha = 0.05$ \n")
9   cat("to conclude that the true mean BP reduction exceeds 10 mmHg.")
10 }
```

Decision: Fail to reject H₀

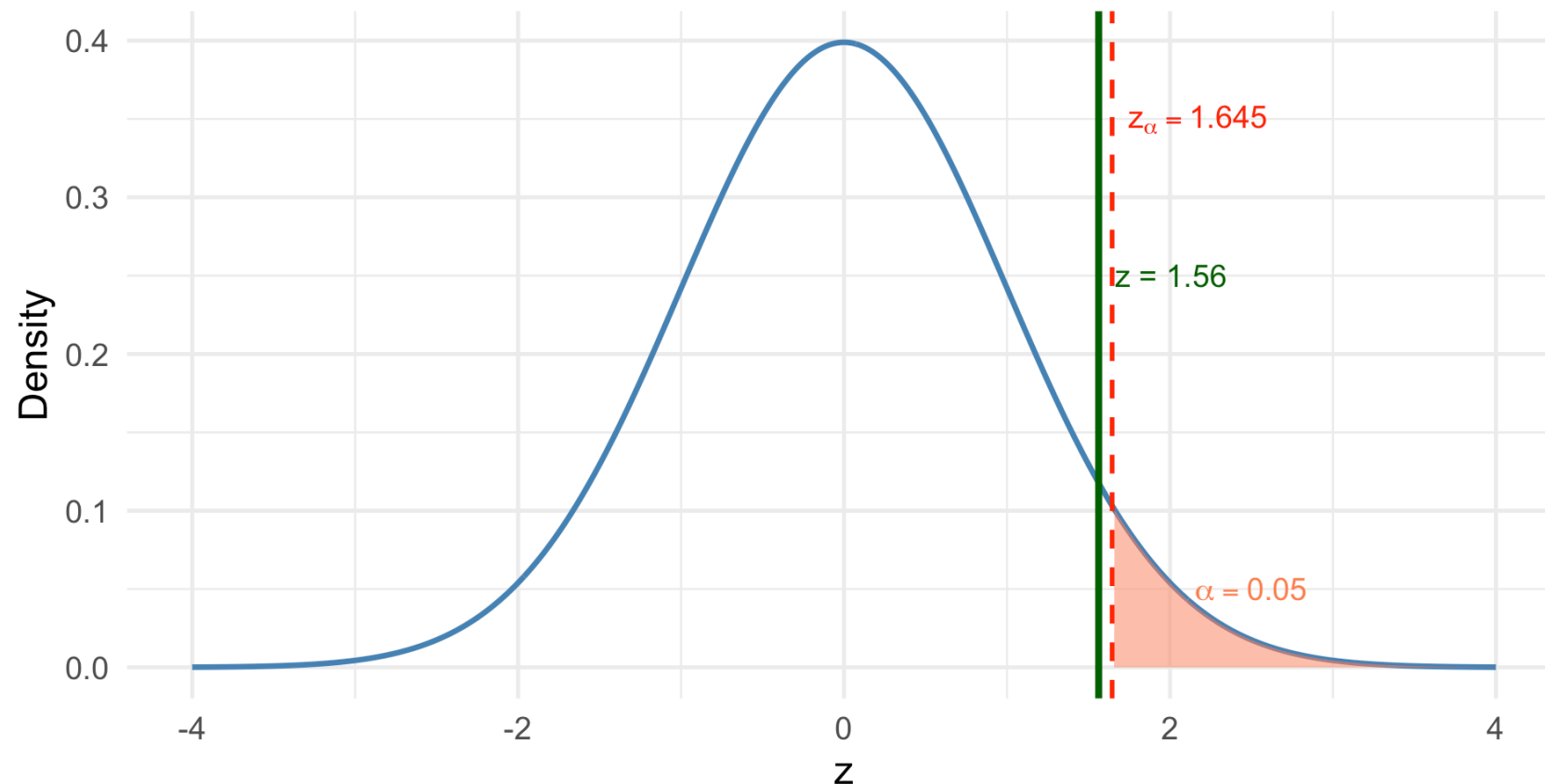
Interpretation: There is insufficient evidence at $\alpha = 0.05$
to conclude that the true mean BP reduction exceeds 10 mmHg.

Visualizing the Test Result

► Code

Hypothesis Test for Blood Pressure Drug

Test statistic falls in rejection region $\rightarrow$ Reject H_0



Error Analysis for This Test

► Code

Type I Error Probability: $\alpha = 0.05$

► Code

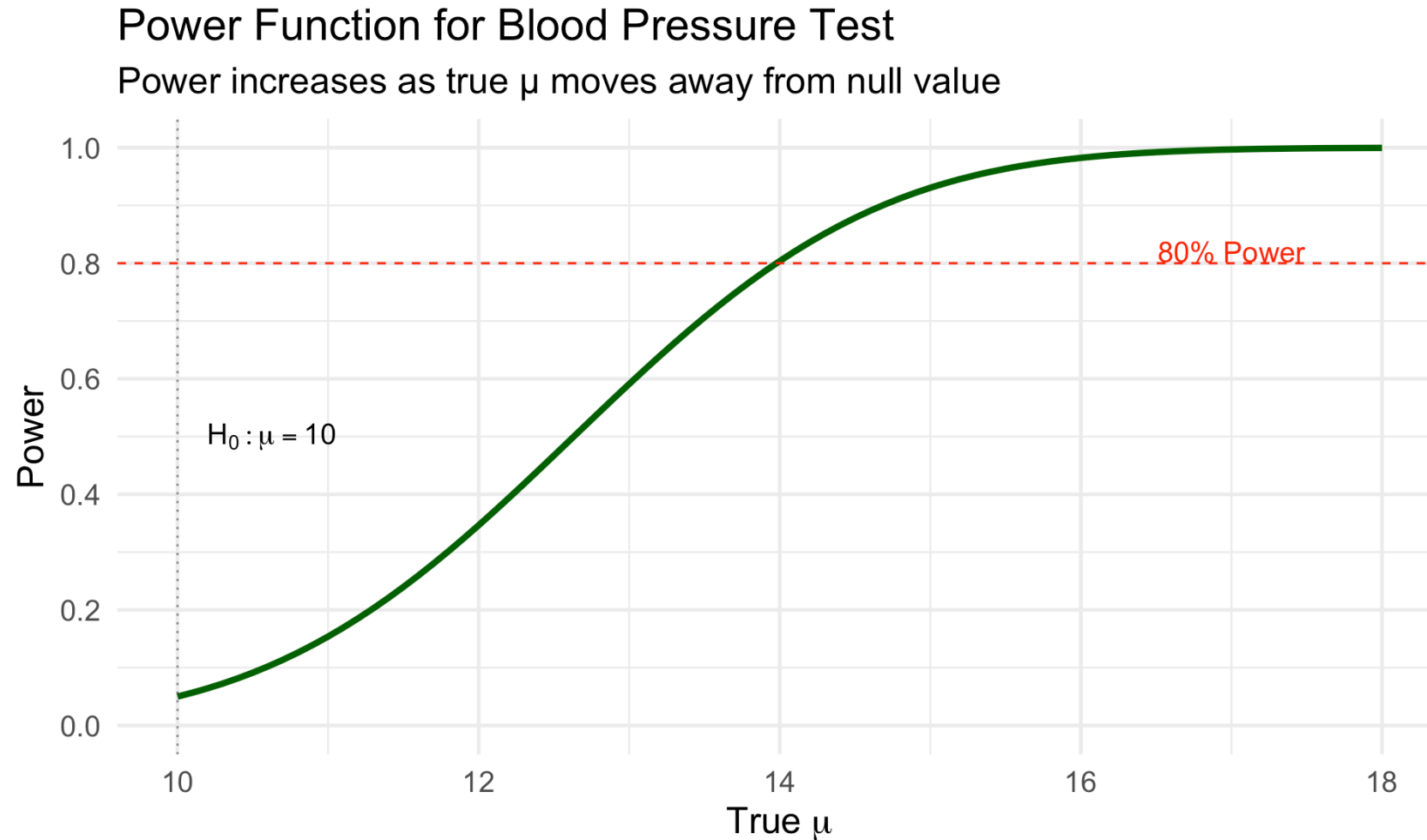
Type II Error and Power for different true μ values:

► Code

```
# A tibble: 5 × 3
  `True  $\mu` `β (Type II)` `Power (1-β)`
    <dbl>      <dbl>      <dbl>
1      11      0.846      0.154
2      12      0.654      0.346
3      13      0.409      0.591
4      14      0.196      0.804
5      15      0.0694     0.931$ 
```

Power Curve

► Code



Connection to Confidence Intervals

CIs and Tests: Two Sides of the Same Coin

! Key Relationship

A $100(1 - \alpha)\%$ confidence interval contains all parameter values that would **not** be rejected by a two-sided test at level α .

- If the CI **excludes** the null value $\theta_0 \rightarrow$ Reject H_0
- If the CI **includes** the null value $\theta_0 \rightarrow$ Fail to reject H_0

Example: Connecting CI and Test

For our blood pressure example:

```
1 # 95% confidence interval (two-sided)
2 ci_lower <- xbar - qnorm(0.975) * se
3 ci_upper <- xbar + qnorm(0.975) * se
4
5 cat("95% CI for  $\mu$ : (", round(ci_lower, 2), ",", round(ci_upper, 2), ")\n")
```

95% CI for μ : (9.36 , 15.64)

```
1 cat("Null value  $\mu_0 = 10$ \n\n")
```

Null value $\mu_0 = 10$

```
1 if (ci_lower > 10) {
2   cat("Since 10 is below the CI, we would reject  $H_0: \mu = 10$ \n")
3   cat("in favor of  $H_a: \mu \neq 10$  at  $\alpha = 0.05$ ")
4 } else if (ci_upper < 10) {
5   cat("Since 10 is above the CI, we would reject  $H_0: \mu = 10$ ")
6 } else {
7   cat("Since 10 is within the CI, we would fail to reject  $H_0: \mu = 10$ \n")
8   cat("at  $\alpha = 0.05$ ")
9 }
```

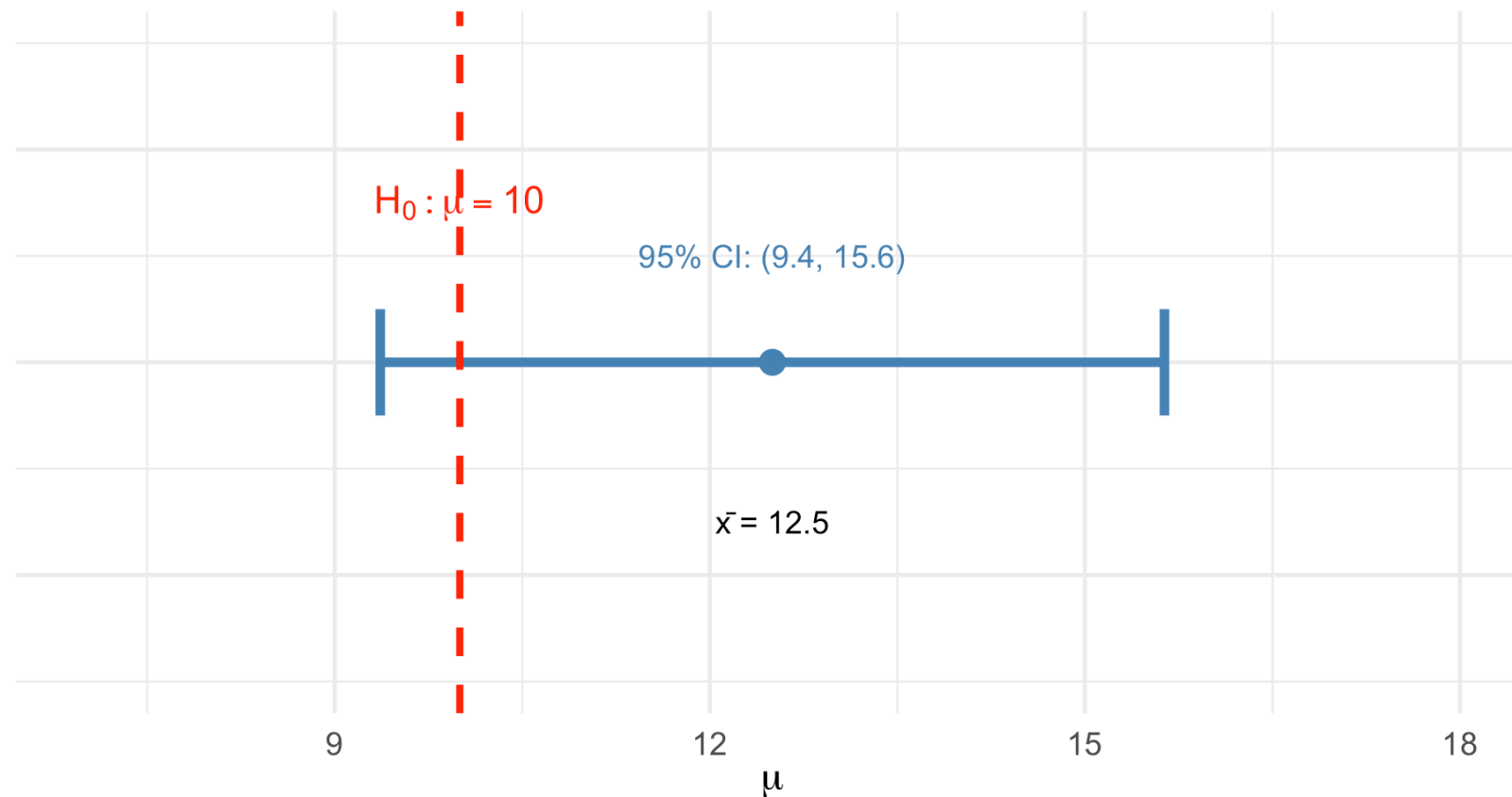
Since 10 is within the CI, we would fail to reject $H_0: \mu = 10$
at $\alpha = 0.05$

CI-Test Duality Visualized

► Code

Confidence Interval and Hypothesis Test

Null value (10) is below the CI → Reject H_0



Summary and Key Takeaways

The Hypothesis Testing Framework

! Core Components

1. **Null Hypothesis (H_0):** The status quo claim (initially assumed true)
2. **Alternative Hypothesis (H_a):** What we want to show (contradicts H_0)
3. **Test Statistic:** Function of sample data for making decisions
4. **Rejection Region:** Values of test statistic leading to rejection
5. **Significance Level (α):** Probability of Type I error

The Two Types of Errors

	H_0 True	H_0 False
Reject H_0	Type I Error (α)	Correct (Power)
Fail to Reject	Correct	Type II Error (β)

The Two Types of Errors

	H_0 True	H_0 False
Reject H_0	Type I Error (α)	Correct (Power)
Fail to Reject	Correct	Type II Error (β)

Key points:

- We control α by choosing it before the test
- β depends on the true parameter value
- Power = $1 - \beta$ = probability of correctly rejecting false H_0
- Tradeoff: Can't minimize both errors without increasing n

Connection to Previous Material

Topic	Lesson	Connection to Testing
Point Estimation	2-4	Test statistics are often MLEs
Sufficient Statistics	5	Best tests use sufficient statistics
Confidence Intervals	7-8	CIs give all non-rejected values
Bootstrap	9-11	Bootstrap hypothesis tests use resampling

Looking Ahead

Coming up in this course:

- **P-values:** Continuous measure of evidence against H_0
- **Power and sample size:** How many subjects do we need?
- **Tests for specific parameters:** Mean (z and t), proportion, variance
- **Two-sample tests:** Comparing populations
- **Bootstrap hypothesis tests:** Resampling-based testing

References

- Devore, Berk, and Carlton. *Modern Mathematical Statistics with Applications* (Springer). Chapter 9.1
- Chihara and Hesterberg. *Mathematical Statistics with Resampling and R* (Wiley). Chapter 3.1-3.2, 8.1-8.2
- Wasserman, L. (2004). *All of Statistics*. Chapter 10.

Questions?

Thank you!